First Semester

21CYB101J - CHEMISTRY

(For the candidates admitted during the academic year 2024-2025)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours Max. Marks: 75

PART - A (20 x 1 = 20 Marks)
Answer **ALL** Questions

		Marks	BL	CO	PO
1.	For a high spin d^4 octahedral complex the crystal field splitting energy will be A) $-0.6\ \Delta_o$ B) $-0.8\ \Delta_o$ C) $0.6\ \Delta_o$ D) $0.8\ \Delta_o$	1	3	1	1
2.	Which of the following is paramagnetic? [Z for Co= 27, Fe=26, Ni= 28] A) [Pt(NH3)3Cl]+ B) [Ni(CN)4]2− C) $[CoBr]_4^{2-}$ D) $[Co(NH_3)_6]^{3+}$	1	2	1	1
3.	The crystal field splitting energy for octahedral and tetrahedral complexes is related as A) $\Delta_t \approx 1/2\ \Delta o$ B) $\Delta o \approx 2\ \Delta t$ C) $\Delta o \approx 4/9\ \Delta t$ D) $\Delta_t \approx 4/9\ \Delta o$	1	2	1	1
4.	Suggest the hard acid from the following ions given: A) Cu^+ B) Mg^{2+} C) Cd^{2+} D) Hg^{2+}	1	2	1	1
5.	According to the convention, the Daniel cell is represented as _____ A) Zn l $ZnSO_4$ll Cu l $CuSO_4$, E = 1.09 volt B) $ZnSO_4$ l Zn ll $CuSO_4$ l Cu, E = 1.09 volt C) Zn l $ZnSO_4$ll $CuSO_4$ l Cu, E = 1.09 volt D) Zn l ZnSll $CuSO_4$ l Cu, E = 1.09 volt	1	2	2	2
	The following are state functions EXCEPT A) enthalpy (H) B) heat (q) C) internal energy(U) D) entropy (S)	1	2	2	2
	When zinc metal, is attached to the ship's hull made of iron, zinc corrode itself instead of the hull. The type of corrosion is an example of ---- A) Galvanic corrosion B) oxidation corrosion C) liquid metal corrosion D) pitting corrosion	1	3	2	2
	f the solubility product of magnesium hydroxide is 2.00×10^{-11} $mol^3\ dm^{-9}$ at 298 k, alculate its solubility in mol dm^{-3} at that temperature.) 1.71×10^4 mol dm^{-3} B) 1.71×10^{-4}mol dm^{-3}) 2.71×10^{-4}mol dm^{-3} D) 3.42×10^{-5}mol dm^{-3}	1	4	2	3

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PART - B (5 x 8 = 40 Marks)
Answer ALL the Questions

21 a. i)	Write a short note on spectrochemical series? Explain the impact of ligands in the formation of high and low spin complexes with suitable examples. [6]	8	2	1	4
ii)	What is Pairing energy (P)? Give the relation between crystal field splitting in octahedral complexes (Δ_O) and pairing energy (P). [2]				
	(OR)				
b. i)	Write a note on the geometrical isomerism exhibited by octahedral complexes.	8	2	1	1
ii)	Aluminium occurs in nature as an oxide, whereas Platinum occurs as sulfide. Explain it on the basis of HSAB principle..	8	2	2	1
22 a.	Derive Nernst equation and give its application towards emf determination by potentiometric titration.	8	2	2	2
	(OR)				
b.	Explain in detail the mechanism with equations involved in dry corrosion.	8	3	3	3
23 a.	Sketch the potential diagram and discuss in detail the conformational analysis of n-butane.	8	3	3	4
	(OR)				
b.	Discuss elements of symmetry with an example for each.	8	1	4	1
24 a. i)	Give the differences between addition and condensation reactions.	8	1	4	1
ii)	Compare isotactic and syndiotactic polymers				
	(OR)				
b.	Explain in detail n and p doping in conducting polymers.	8	2	5	1
25 a.	Explain the various regions in stress-strain graph with a neat diagram.	8	2	5	3
	(OR)				
b.	Discuss on ceramic and metal matrix composites with examples.	8	2	5	3

PART - C (1 x 15 = 15 Marks)
Answer ANY ONE Questions

		Marks	BL	CO	PO
26	Discuss in detail Pourbaix diagram of Iron system with a neat sketch.	15	4	2	3
27	a. What is the principle underlying in XPS? Give the merits and demerits of XPS analysis [8]	15	2	5	5
	b. What is Bragg's Law? Prove $n\lambda = 2d \sin\theta$. [7]				

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